PHASE 2: REQUIREMENT ANALYSIS

EV Market Performance & Insights

Date	*20 – October - 2026*
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Project Name	*Electric vehicle Market Performance & Insights*

Customer Journey Map • Solution Requirement • Data Flow Diagram • Technology Stack

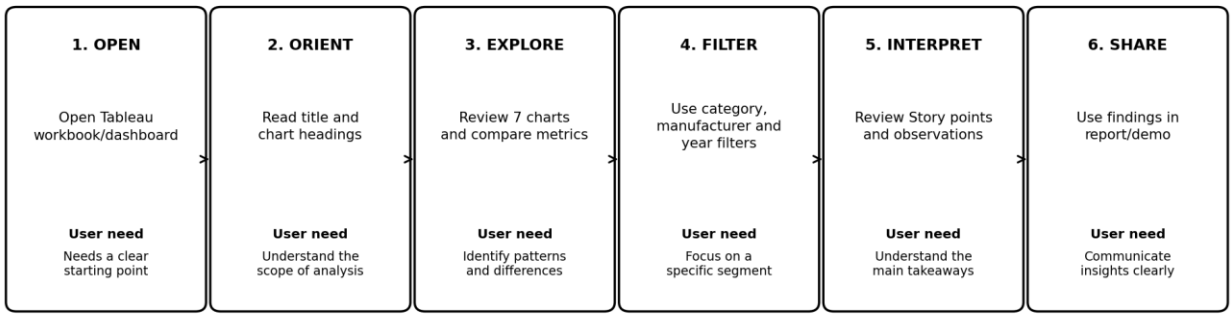
Purpose: This document defines the intended user journey, requirements, data flow, and tools for the Tableau-based EV market analysis project. It is prepared around the visible dashboard and project assets; confirm any implementation-specific details before final submission.

1. Customer Journey Map

Persona: an analyst or business user who wants to explore EV market patterns using the available dataset. This journey is an illustrative planning model, not a result of a formal user study.

Stage	User Action	User Need / Question	Design Opportunity
Open	Open the Tableau workbook/dashboard.	Where should I start?	Clear title and dashboard overview.
Orient	Read chart titles and understand the measures.	What does each chart show?	Descriptive chart titles and readable axes.
Explore	Compare the seven visualizations.	Which categories or manufacturers differ?	Consistent formatting and appropriate chart types.
Filter	Select category, manufacturer, or model year where available.	Can I focus on one segment?	Visible, functioning filters with clear selections.
Interpret	Review patterns and the three Story points.	What are the key observations?	Use verified values and avoid unsupported claims.
Share	Use the dashboard in the report or demo.	How can I communicate the result?	Readable layout and concise takeaway statements.

CUSTOMER JOURNEY MAP — EV MARKET ANALYSIS USER



Potential friction points: crowded labels • unclear metric definitions • filters that are hard to notice • unsupported conclusions

Design response: readable titles and labels • relevant filters • consistent measures • findings grounded in the dataset

Potential Pain Points and Responses

- Crowded labels can make categories difficult to read; adjust label density, chart size, or sorting.
- Unclear measures can lead to misinterpretation; use explicit chart titles and axis labels.
- Filters may be missed or confusing; group them logically and test their behavior.
- Findings should be based on visible data; verify actual leaders, values, and trends before reporting.

2. Solution Requirement

The solution is a Tableau dashboard and Story for descriptive exploration of the EV dataset. Requirements below describe the intended behavior and quality criteria for the current project scope.

2.1 Functional Requirements

ID	Requirement	Acceptance Check
FR-01	Load the available EV dataset into Tableau.	Relevant fields are visible in the Tableau data pane/data source.
FR-02	Display seven visualizations for the selected EV analysis areas.	All seven charts appear and show data.
FR-03	Compare annual sales across model years.	Annual sales trend chart uses model year and the applicable sales measure.
FR-04	Compare manufacturers by annual sales units.	Manufacturer chart displays the selected manufacturers and values.
FR-05	Show category and battery technology distributions.	The corresponding charts display category/technology comparisons.
FR-06	Compare average driving range and customer ratings by EV category.	Range and rating charts use the relevant measures and category dimension.
FR-07	Provide available interactive filters.	Category, manufacturer, and model-year filters respond as configured.
FR-08	Organize the analysis into three Tableau Story points.	Story navigation includes EV Market Overview, Vehicle & Battery Insights, and Manufacturer & Country Insights.
FR-09	Support communication of findings.	Report and demo summarize only findings confirmed by the dashboard.

2.2 Non-Functional Requirements

- Usability: chart titles, labels, and filters should be easy to understand.
- Readability: labels should not overlap unnecessarily and text should remain legible at normal screen size.
- Consistency: use consistent number formats, colors, and naming across related charts.
- Accuracy: displayed values should correspond to the connected dataset and selected aggregations.
- Performance: workbook navigation and filter interactions should remain reasonably responsive on the intended computer.
- Maintainability: use meaningful worksheet, dashboard, and Story point names.

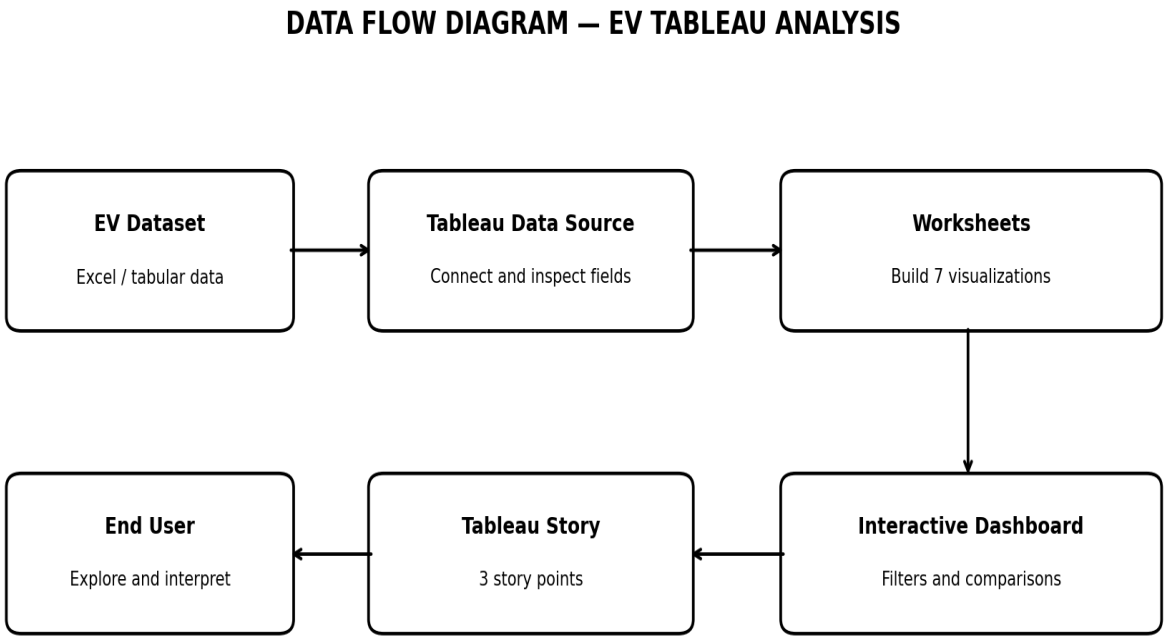
2.3 Scope and Limitations

- In scope: descriptive analysis of the available dataset using Tableau visualizations, filters, and Story points.

- Out of scope unless separately implemented: live data integration, predictive modelling, and automated market updates.
- Data limitations: the findings represent the dataset provided and should not automatically be generalized to the entire EV market.

3. Data Flow Diagram (DFD)

The diagram below summarizes the current analytical workflow from the available dataset to Tableau worksheets, dashboard, Story, and end-user interpretation.



Scope note: the diagram represents the current Tableau workflow; it does not imply a live data feed or predictive model.

3.1 Data Flow Explanation

- Input: the EV dataset files available to the project team.
- Data connection: Tableau connects to the relevant tabular data and exposes dimensions and measures.
- Visualization: worksheets use fields and aggregations to build the seven charts.
- Dashboard: worksheets are combined into one view with the configured filters.
- Story: the analysis is arranged into three points to communicate related views in sequence.
- Output: users explore and interpret descriptive patterns and use verified observations in the project report.

Note: This is a high-level DFD for the current Tableau workflow. It does not claim that a separate ETL pipeline, database server, or live data source was implemented.

4. Technology Stack

The table distinguishes tools visible in the project assets from optional tools that should only be included if actually used.

Layer	Technology / Tool	Purpose / Status
Data files	Excel-compatible dataset files (ev_market_train, ev_market_test)	Visible in the project folder; used as project data assets. Confirm which file is connected to the final workbook.
Data visualization	Tableau (packaged workbook .twbx)	Core platform for worksheets, dashboard, filters, and Story.
Visualization layer	Seven Tableau charts	EV category, annual sales, manufacturer, battery technology, range, customer ratings, and country distribution.
Interactivity	Tableau filters	Available dashboard filters include category, manufacturer, and model year, as shown in the dashboard screenshot.
Storytelling	Tableau Story	Three story points organize the overview, vehicle/battery, and manufacturer/country views.
Documentation	Microsoft Word and PDF export	Prepare project reports and convert final submissions to PDF as required.
Demo	Screen recording tool	To be selected/confirmed for the project demonstration video.

